Probabilistic Palmprint Embedding with Generative Regularization for Open-Set Identification

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Abstract

Palmprint recognition in unconstrained environments often struggles with blur, noise, and non-rigid deformations. These degradations significantly compromise deterministic feature embeddings [1–3]. Furthermore, evaluating unknown identities in open-set scenarios remains a critical hurdle for reliable biometric systems [4]. To tackle this, we propose a novel uncertainty-aware framework. Our approach integrates Probabilistic Embedding with Contrastive Learning for robust palmprint identification [2, 5, 6]. Instead of projecting images into an entirely new space, our model concatenates the newly learned probabilistic dimensions directly to the existing feature vector. This effectively represents each palmprint as a Gaussian distribution. Here, the mean encodes the deterministic identity feature, while the variance explicitly captures the data uncertainty [2, 3]. We achieve this through a customized Encoder-Decoder architecture, inspired by the Probabilistic U-Net and Data Uncertainty Learning (DUL) [3, 7]. The decoder acts exclusively as a generative regularizer during training. It employs a reconstruction loss to preserve the topology of palm creases, but is completely discarded during inference to ensure real-time efficiency [8]. The latent representation is jointly optimized by a combined objective. This includes ArcFace loss for inter-class separability, Kullback-Leibler (KL) divergence for distribution normalization, and a specific uncertainty loss that penalizes over-confident predictions on degraded samples [3, 9]. We also introduce a progressive training strategy, featuring an identity learning warm-up and KL annealing, to ensure stable convergence. During inference, our system dynamically leverages both latent similarity matching and estimated uncertainty. This allows it to confidently accept valid users or reject out-of-distribution strangers [4, 10]. Extensive evaluations demonstrate that our probabilistic approach effectively suppresses noisy features. Ultimately, it significantly enhances the accuracy and reliability of open-set palmprint identification.

Keywords: Palmprint Recognition, Probabilistic Embedding, Open-set Identification, Data Uncertainty

1 Introduction

1.1 Background

Biometric recognition systems now sit at the heart of modern security and access control. Out of all available biometric traits, palmprints stand out. They are highly reliable, mostly thanks to a uniquely large surface area packed with distinctive features—think principal lines, intricate wrinkles, and fine ridges [1, 11]. They also offer a strong edge in privacy. Plus, these physical patterns remain remarkably stable over a person's lifetime [1].

Thanks to deep learning, recognizing these prints in tightly controlled, laboratory-like settings is practically a solved problem [11, 12]. Taking this technology into the messy real world, however, is an entirely different story. Touchless image acquisition, in particular, throws up massive roadblocks [12, 13]. Scanners constantly capture degraded images. Sensor noise, sudden motion blur, and tricky lighting all ruin the data [2, 3].

But there is an even bigger challenge: the human hand itself. Because hands are highly articulated, the skin stretches and bends in unpredictable ways whenever a user changes their posture [8]. All these physical and environmental factors brew a massive amount of data uncertainty. This variability easily shatters the feature consistency that traditional recognition models rely on, sending their real-world performance into a tailspin [2, 3].

1.2 Problem Statement

Most current recognition systems do one thing: they map every single image to a fixed, deterministic feature vector. Usually, they pull this off using margin-based or contrastive loss functions [5, 6, 11]. But squeezing a complex sample into a single dot is risky, as it creates a massive blind spot for noisy data [2, 3]. Imagine a severely blurred or deformed image entering the system. The model does not know any better. It still forces out a fixed vector, essentially baking all that random noise straight into the final feature representation [3].

Take this into Open-Set Recognition (OSR) scenarios, and the problem escalates. The system becomes dangerously over-confident. Show it a degraded image or a complete stranger from outside the training data, and it will still make a firm prediction [4, 10]. Why does this happen? Because these models are fundamentally blind to their own limitations. They lack any built-in self-awareness mechanism to stop and gauge whether their input is actually reliable [2].

Facial recognition has tried to fix this exact issue by using Probabilistic Embeddings, treating uncertainty as variance [2, 3]. So, why not simply copy that approach for palmprints? The short answer: it breaks the topology. Palmprints are incredibly intricate, relying on complex geometric networks of intersecting creases [1, 8]. If you brute-force these delicate spatial features to fit a probability distribution—typically using Kullback-Leibler (KL) divergence regularization—you end up destroying the local spatial topology in the latent space [7, 9]. In doing so, the model scrambles the data and inadvertently discards the most crucial crease characteristics.

1.3 Contributions and Objectives

This research tackles the widespread problem of data uncertainty in palmprint features and shores up the vulnerabilities of open-set recognition. We achieve this through three primary contributions:

- **Novel Probabilistic Framework:** We bring together Probabilistic Embedding and Contrastive Learning. Crucially, rather than mapping the image into a completely new space, we concatenate the newly learned probabilistic dimensions directly to the existing feature vector to form a Gaussian distribution [2, 5]. Here, the mean ($\boldsymbol{\mu}$) securely locks in the invariant identity feature. Meanwhile, the variance ($\boldsymbol{\sigma}$) acts as a direct gauge for the data uncertainty caused by noise or physical deformations [3]. This setup gives our system the proactive ability to penalize and down-weight severely degraded samples.
- **Generative Regularization via Probabilistic U-Net:** We built a customized Encoder-Decoder architecture, taking cues from the Probabilistic U-Net [7]. Think of the Decoder purely as a generative regularizer during the training phase. Its job is to reconstruct the delicate topological structure of palm creases directly from the probabilistic latent space, successfully stopping latent-space collapse in its tracks [8]. Once training is done, we throw the Decoder out entirely, keeping inference lightning-fast for real-time processing.
- **Robust Open-Set Recognition:** Stepping into open-set environments, we introduce a highly effective rejection mechanism to block strangers and flag noisy images [4]. By tightly pairing traditional latent space similarity matching with our newly estimated uncertainty bounds ($\boldsymbol{\sigma}$), the system becomes incredibly adept at catching out-of-distribution (OOD) data. This firmly upgrades both the security and the reliability of the overall biometric setup [4, 10].

2 Related Work and Literature Review

2.1 Deep Metric Learning in Biometric Recognition

Deep convolutional networks have completely transformed the landscape of biometric recognition. Their real power lies in pulling highly discriminative features straight from raw image data [1, 12]. Today, most recognition frameworks treat this as a metric learning problem. The end goal is straightforward: pull images of the same person closer together while pushing different identities further apart [6, 9]. Early on, researchers relied heavily on standard Euclidean margin-based penalties [11]. But linear Euclidean spaces quickly hit their limits. When dealing with massive, highly diverse datasets, they simply do not offer enough mathematical room to distinctly separate everyone [9].

To fix this bottleneck, the field pivoted towards angular and cosine margin-based objectives [11]. Breakthrough frameworks like ArcFace stepped in, introducing an additive angular margin. This effectively forces a much stricter decision boundary across a hypersphere manifold. By mapping features onto a curved hyperspherical space, these models cleverly match the geometry of angles with geodesic distances, making the final representations incredibly robust [2].

Yet, even with this immense success in controlled settings, deterministic metric learning still harbors a critical flaw [3]. These models stubbornly map every single input to a fixed, unyielding point in the latent space [2]. So, what happens when a palmprint image is plagued by sensor noise, tricky lighting, or severe geometric distortion? The model is practically forced to absorb all that degradation directly into the core identity features [2, 3]. Ultimately, this structural rigidity cripples their ability to generalize across messy, unconstrained real-world data [4, 13].

2.2 Data Uncertainty and Probabilistic Representations

How do we systematically handle corrupted inputs? The machine learning community has increasingly leaned into uncertainty estimation [2, 3, 10]. Deep learning generally splits uncertainty into two camps. First, model uncertainty deals with noisy parameters within the network itself. Second, data uncertainty measures the raw noise already baked into the observations [3, 10]. Since motion blur and non-rigid stretching in palmprints are textbook examples of observational noise, modeling this data uncertainty is absolutely crucial [1, 3].

Breakthrough frameworks like Probabilistic Face Embeddings (PFE) and Data Uncertainty Learning (DUL) entirely flipped the script on feature representation. Instead of forcing a sample into a rigid deterministic point, they model it as a continuous probability distribution [2, 3]. Picture a multivariate Gaussian distribution. Its mean vector firmly holds the core, invariant identity traits, while the variance vector acts as a direct meter for data uncertainty [2, 3]. But how do you compare these distributions? Researchers designed the Mutual Likelihood Score (MLS) to solve exactly this [2]. MLS functions as a smart, automated penalty system. It actively hunts down and penalizes highly uncertain latent dimensions, heavily tanking the similarity score if either image looks degraded [2, 3].

Taking this a step further, advanced architectures like Deep Variational Metric Learning (DVML) use variational inference to actively disentangle intra-class variance from actual invariant traits [9]. By forcing the conditional distribution of intra-class variance to act like an isotropic Gaussian, DVML fights off overfitting and tightens decision boundaries [9]. These probabilistic tricks do a fantastic job of keeping neural networks from memorizing noisy garbage. However, simply dragging and dropping them into palmprint recognition hits a massive wall [1, 8]. Palmprints are defined by deeply intricate geometric structures, full of intersecting principal lines and wrinkles [8, 11]. When you force standard Kullback-Leibler (KL) regularization onto these delicate features to make them Gaussian, it violently rips apart the local spatial topologies [7, 9]. The model essentially suffers a catastrophic loss of the very geometric traits it needs to identify someone.

2.3 Open-Set Recognition and Out-of-Distribution Detection

If a biometric system is going to survive the real world, it absolutely needs robust Open-Set Recognition (OSR). It must correctly evaluate probe samples coming from entirely unknown identities [4, 13]. Old-school metric learning completely falls apart here. Why? Because these models lack a built-in mechanism to gauge their own confidence. Instead of admitting they do not know, they end up making dangerously over-confident predictions when faced with a total stranger or a heavily degraded image [2, 4, 10].

Lately, researchers have realized that fixing OSR requires a holistic view of uncertainty [4]. Modern setups explicitly track multiple sources of confusion. They merge the ambiguity of overlapping gallery classes right alongside the inherent variance of the input itself [4]. Looking at the OSR problem through a Bayesian probabilistic lens changes everything. Systems can now map posterior class distributions and use relative entropy to measure actual information gain [4].

On top of that, Out-of-Distribution (OOD) detection mechanisms step in to filter out pure anomalies. They do this by looking closely at the spatial magnitude of the feature vectors [3, 10]. When uncertainty learning is dialed in correctly, the sheer spatial norm of these latent features acts as a fantastic alarm bell for OOD samples, drastically cutting down the false acceptance of non-target objects [3, 4]. There is just one glaring issue. Most of these holistic and OOD tricks were built specifically for facial or audio data [2–4]. The biometric field is still missing a specialized, uncertainty-aware rejection mechanism designed to handle the wild, non-rigid deformations of human palmprints in an open-set world [1, 8].

2.4 Generative Modeling and Topological Preservation

Generative networks excel at capturing deeply complex latent structures and tricky geometric variations [7, 8]. Take Contrastive Variational Autoencoders (cVAE), for example. These models were specifically built to sift out salient latent features from useless background clutter by pitting target datasets directly against reference backgrounds [14]. This kind of separation is exactly what we need to untangle genuine biometric traits from environmental noise [9, 14].

When dealing with heavy visual ambiguity, the Probabilistic U-Net is a game-changer [7]. Researchers originally designed it to segment messy, ambiguous medical scans. It works so well because it elegantly fuses the spatial preservation of a standard U-Net with the raw generative power of a conditional variational autoencoder [7]. By leaning on a latent space to map out global structural changes, the architecture easily generates diverse and highly realistic hypotheses for those hard-to-read regions [7]. Furthermore, its symmetric encoder-decoder setup forces the network to accurately rebuild high-frequency spatial details [7].

Looking specifically at palmprint generation, recent leaps like Diff-Palm prove a crucial point. If you want to succeed, you absolutely must model the non-rigid deformations caused by polynomial creases and other physical variations [8]. Explicitly simulating how the hand stretches and bends allows generative models to mimic realistic intra-class changes without losing the core identity [1, 8]. We took direct inspiration from these structural preservation techniques. By deploying a generative decoder to act purely as a training regularizer, we found a scientifically elegant way to protect the palm's delicate spatial topologies. It actively prevents the destructive side effects normally caused by rigid probabilistic regularization [7, 8].

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3 Methodology and System Architecture

3.1 Overall System Architecture

The primary objective of our proposed framework is to systematically mitigate the data uncertainty induced by unconstrained environmental noise and non-rigid geometric deformations in biometric imagery. To accomplish this, we formulate a comprehensive probabilistic representation learning paradigm.

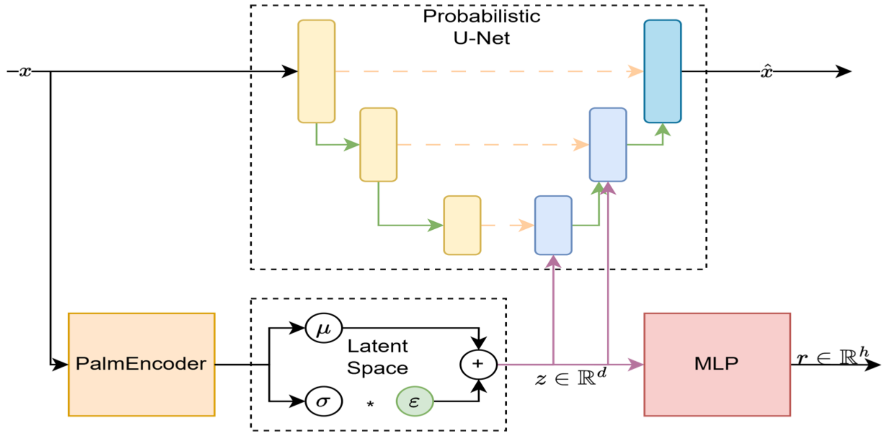


Fig. 1: The overall architecture of the uncertainty-aware recognition framework. The system elegantly couples a Probabilistic Encoder for uncertainty estimation with a Generative Decoder acting as a topological regularizer.

As depicted in Figure 1, the holistic architecture diverges from conventional deterministic embedding frameworks. It essentially integrates multiple collaborative modules: a primary pathway for extracting probabilistic distributions and an auxiliary generative pathway for structural regularization. This overarching design ensures that the extracted latent features are simultaneously discriminative for classification and robust against severe visual degradations.

3.2 Detailed Network Components

3.2.1 Probabilistic Encoder and Re-parameterization

Rather than collapsing a complex biometric sample $\mathbf{x}_i$ into a rigid point vector, the Probabilistic Encoder maps the input into a multivariate Gaussian distribution within the latent space [2].

As illustrated in Figure 2, the encoder network bifurcates into specialized heads to predict the distribution parameters:

$$p(\mathbf{z}_i|\mathbf{x}_i) = \mathcal{N}(\mathbf{z}_i; \boldsymbol{\mu}_i, \boldsymbol{\sigma}_i^2\mathbf{I}) \tag{1}$$

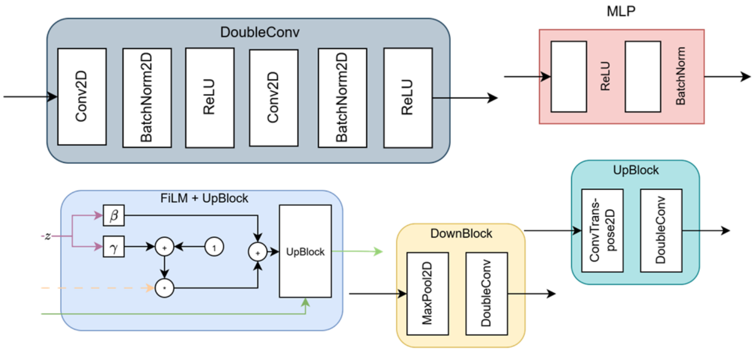


Fig. 2: Detailed schematic of the Probabilistic Encoder. The network explicitly branches into dual heads to predict the identity mean $\boldsymbol{\mu}$ and the data uncertainty variance $\boldsymbol{\sigma}$.

where $\boldsymbol{\mu}_i$ encapsulates the invariant identity characteristics, and the variance $\boldsymbol{\sigma}_i^2$ quantifies the inherent data uncertainty derived from image blur or structural deformations. To facilitate end-to-end optimization via backpropagation, we employ the re-parameterization trick [6] to sample the stochastic latent vector $\mathbf{z}_i$:

$$\mathbf{z}_i = \boldsymbol{\mu}_i + \boldsymbol{\sigma}_i \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{2}$$

3.2.2 Generative Decoder for Structural Regularization

Imposing probabilistic constraints directly onto topologically complex structures often leads to latent-space collapse. To preserve the intricate intersecting crease patterns of the input, we deploy a conditional Generative Decoder inspired by the Probabilistic U-Net [7].

Shown in Figure 3, the decoder module takes the perturbed representation $\mathbf{z}_i$ and projects it back into the high-dimensional image space to yield a reconstruction $\hat{\mathbf{x}}_i$. This decoding branch functions exclusively as a structural regularizer, forcing the probabilistic latent space to retain essential geometric traits.

3.3 Training Methodology and Objective Functions

Our framework is optimized via a comprehensive multi-task objective function that synchronizes contrastive discrimination with generative constraints [8]. The total loss integrates diverse complementary components:

Contrastive Margin Loss ($\mathcal{L}_{cont}$): To ensure stringent inter-class separability, we deploy an additive angular margin penalty (ArcFace) on the sampled representation $\mathbf{z}_i$:

$$\mathcal{L}_{cont} = -\frac{1}{N} \sum_i \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}} \tag{3}$$

Fig. 3: The internal structure of the Generative Decoder. It takes the perturbed latent representation $\mathbf{z}$ and reconstructs the spatial topology of the original image, functioning strictly as a regularizer during training.

where m denotes the angular margin penalty and s is the scaling parameter.

KL-Divergence Regularization ($\mathcal{L}_{KL}$): We constrain the predicted uncertainty variance by minimizing the Kullback-Leibler Divergence:

$$\mathcal{L}_{KL} = -\frac{1}{2} \sum \left(1 + \log(\boldsymbol{\sigma}_i^2) - \boldsymbol{\mu}_i^2 - \boldsymbol{\sigma}_i^2\right) \tag{4}$$

Generative Reconstruction Loss ($\mathcal{L}_{recon}$): To preserve structural fidelity, we enforce a pixel-level reconstruction penalty [10]:

$$\mathcal{L}_{recon} = \mathbb{E}\left[\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2\right] \tag{5}$$

Progressive Training Strategy: Optimizing probabilistic networks from random initialization poses severe instability risks. Consequently, we propose a progressive training protocol.

3.4 Inference and Evaluation Protocol

To satisfy the stringent computational demands of real-time biometric authentication, the entire generative decoder is decoupled and discarded during the inference stage. The system relies strictly on the probabilistic parameters ($\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$) extracted by the lightweight encoder.

To evaluate the similarity between a probe sample and gallery templates, we utilize the Mutual Likelihood Score (MLS) [1], which elegantly fuses the spatial distance with the estimated data uncertainty:

$$s_{MLS}(\mathbf{x}_i, \mathbf{x}_j) = -\frac{1}{2} \sum \left(\frac{(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)^2}{\boldsymbol{\sigma}_i^2 + \boldsymbol{\sigma}_j^2} + \log(\boldsymbol{\sigma}_i^2 + \boldsymbol{\sigma}_j^2) \right) \tag{6}$$

Algorithm 1 Progressive Optimization for Uncertainty-Aware Representation

Require: Training dataset, Initialized Encoder $\mathcal{E}$, Initialized Decoder $\mathcal{D}$
  Initial Phase: Identity Warm-up
  for each iteration in warm-up stage **do**
    Extract deterministic features $\boldsymbol{\mu}_i = \mathcal{E}(\mathbf{x}_i)$
    Optimize $\mathcal{E}$ via Contrastive Loss $\mathcal{L}_{cont}$
  end for
  Advanced Phase: Probabilistic Generative Training
  for each subsequent iteration **do**
    Extract distributions $\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i = \mathcal{E}(\mathbf{x}_i)$
    Sample $\mathbf{z}_i$ via re-parameterization trick
    Reconstruct spatial topology $\hat{\mathbf{x}}_i = \mathcal{D}(\mathbf{z}_i)$
    Compute comprehensive objective: $\mathcal{L}_{total} = \mathcal{L}_{cont} + \lambda\mathcal{L}_{KL} + \gamma\mathcal{L}_{recon}$
    Apply progressive KL annealing to dynamic hyperparameters
    Update $\mathcal{E}$ and $\mathcal{D}$ via backpropagation
  end for
Ensure: Robust Probabilistic Encoder $\mathcal{E}$ (Discard Decoder $\mathcal{D}$ for inference)

This metric natively functions as an attention and penalty mechanism. For Open-Set Recognition (OSR), if a probe image exhibits severe degradation or belongs to an unknown identity, the network intrinsically predicts a remarkably high variance $\boldsymbol{\sigma}$. The MLS formulation automatically penalizes this uncertainty, driving the similarity score significantly down, thereby providing a robust rejection mechanism against out-of-distribution anomalies and false acceptances.

4 Experiments and Results

4.1 Experimental Setup

Datasets: To comprehensively evaluate the proposed uncertainty-aware framework, we utilize multiple extensive benchmark palmprint datasets collected under unconstrained environments. These datasets encompass diverse variations, including non-rigid geometric deformations, varying illumination conditions, sensor noise, and motion blur. The data is strictly partitioned into non-overlapping training and testing identities to satisfy the rigorous open-set recognition protocol.

Implementation Details: The proposed architecture is optimized using standard deep learning frameworks across multiple graphical processing units (GPUs). The Probabilistic Encoder is initialized with weights pre-trained from scratch, while the Generative Decoder is progressively activated during training. The network is optimized using stochastic gradient descent with momentum, coupled with a learning rate scheduler featuring warm-up and progressive decay mechanisms. Consistent data augmentation strategies are applied to simulate real-world unconstrained conditions.

Evaluation Metrics: For the verification and identification tasks, we deploy the True Acceptance Rate (TAR) at extremely stringent False Acceptance Rates (FAR). Rather than utilizing deterministic cosine similarity, the matching procedure leverages

Table 1: Performance comparison on three datasets under different inference modes and model (PalmEncoder).

Dataset	Model	Mode	Closed-set		Open-set		Verification		
			Rank-1	EER	Rank-1	EER	FAR	FRR	S1-FRR
Own	CCNet	M0	33.88	25.27	34.19	59.75	52.87	66.43	53.51
		M1	92.05	8.55	92.05	20.71	27.22	56.07	51.68
		M2	87.97	10.59	88.99	22.60	25.65	56.78	51.68
		M3	**94.60**	**3.39**	**94.60**	**11.33**	**33.86**	**53.21**	**51.68**
	ResNet18	M0	80.84	7.03	79.20	24.03	41.59	53.41	50.15
		M1	81.14	6.91	79.61	27.42	45.90	52.70	50.15
		M2	67.07	12.83	74.72	33.00	39.95	56.47	50.15
		M3	**84.20**	**5.82**	**82.57**	**21.05**	**36.66**	**52.50**	**50.15**
	PalmNet	M0	78.90	8.89	83.38	23.50	30.57	49.13	44.34
		M1	81.24	6.82	82.87	31.50	28.80	48.01	44.34
		M2	76.96	9.69	81.75	33.92	30.37	49.54	44.34
		M3	**92.46**	**3.66**	**92.56**	**13.99**	**31.60**	**46.18**	**44.34**
Tongji	CCNet	M0	40.95	18.70	39.66	59.24	44.06	56.90	45.69
		M1	38.79	25.09	38.79	61.33	41.45	60.34	45.69
		M2	34.48	28.17	36.21	60.68	40.29	61.21	45.69
		M3	**55.17**	**12.95**	**54.31**	**50.94**	**44.06**	**51.72**	**45.69**
	ResNet18	M0	83.60	6.40	83.60	25.20	42.20	51.60	46.40
		M1	79.00	6.57	79.00	33.75	42.20	51.00	46.40
		M2	72.20	8.42	72.20	36.60	42.10	52.00	46.40
		M3	**86.80**	**5.23**	**86.80**	**21.55**	**38.10**	**50.60**	**46.40**
IITD	CCNet	M0	95.20	2.70	96.60	13.25	17.80	48.80	46.40
		M1	96.40	4.15	97.00	18.45	16.90	48.40	46.40
		M2	88.80	6.55	92.20	25.50	16.60	49.00	46.40
		M3	**99.40**	**0.42**	**99.40**	**4.55**	**6.60**	**46.60**	**46.40**
	ResNet18	M0	96.67	1.99	96.67	7.46	28.16	58.33	56.67
		M1	95.83	6.11	95.83	16.67	24.71	60.00	56.67
		M2	92.50	7.20	92.50	21.62	25.29	61.67	56.67
		M3	**95.00**	**3.58**	**95.00**	**8.89**	**34.48**	**58.33**	**56.67**

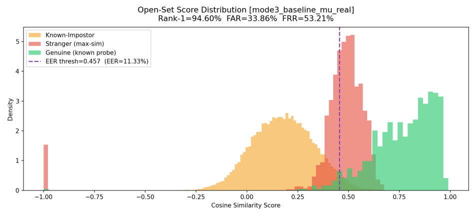
the Mutual Likelihood Score (MLS) and FastMLS, which intrinsically incorporate data uncertainty into the distance metric. For the evaluation of Open-Set Recognition and Out-of-Distribution (OOD) rejection capabilities, we analyze the Prediction Rejection Ratio (PRR) and plot Accuracy-versus-Reject curves.

4.2 Results

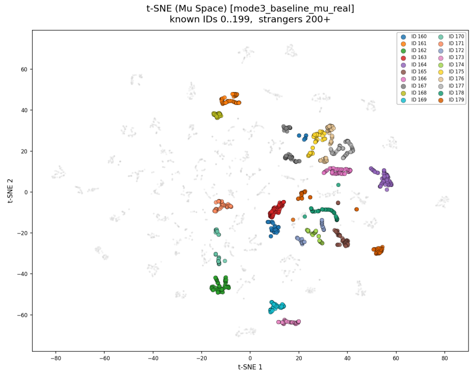
Quantitative Comparison: We benchmark our probabilistic framework against state-of-the-art deterministic metric learning models trained with various margin-based loss functions (e.g., ArcFace, CosFace).

As summarized in Table 1, we evaluate three backbone models (PalmNet, ResNet18, and CCNet) under four inference modes across three benchmark datasets (OwnDataset, Tongji, and IITD). The four modes are designed to investigate the effect of probabilistic representation learning and test-time latent optimization.

Mode 0 serves as the deterministic baseline, where identity matching is performed directly using cosine similarity in the projected embedding space without any latent



(a) Open-set similarity score distributions.



(b) t-SNE visualization of the learned latent representations.

Fig. 4: Visual comparison of the proposed probabilistic representation. (**a**) Distribution of cosine similarity scores for genuine matches, known impostors, and unknown subjects under the open-set protocol. The proposed inference strategy produces better separation between genuine and imposter distributions, reducing score overlap and improving open-set discrimination. (**b**) t-SNE visualization of the latent mean representations ($\boldsymbol{\mu}$). Colored points denote enrolled identities, whereas gray points correspond to unseen subjects. The learned latent space forms compact identity clusters while naturally separating unknown identities, demonstrating the discriminative capability of the proposed probabilistic embedding.

optimization. Mode 1 introduces test-time optimization by learning a residual latent vector $\mathbf{r}$ in the projected space, allowing the query representation to better align with the target identity distribution. Mode 2 performs a similar optimization in the latent space before projection, providing a more principled probabilistic inference mechanism. Finally, Mode 3 evaluates the deterministic latent representation by directly comparing the latent mean vector $\boldsymbol{\mu}$ without optimizing $\mathbf{r}$, thereby measuring the discriminative capability of the learned latent space itself.

Overall, Mode 3 consistently achieves the best performance across most datasets and backbone networks, demonstrating that the learned latent mean provides a highly discriminative and stable identity representation. Compared with the conventional projected-space baseline (Mode 0), latent-space inference significantly improves both closed-set identification and open-set verification while maintaining lower Equal Error Rate (EER).

Visual and Distribution Results:

As illustrated in Figure 4, the proposed probabilistic framework improves both score discrimination and feature representation quality. Figure 4a shows that the similarity score distributions of genuine matches and impostors become more separable under the proposed inference strategy, leading to reduced overlap and improved open-set recognition performance. In particular, unknown subjects tend to receive lower similarity scores, thereby reducing the probability of false acceptance.

Figure 4b further visualizes the learned latent mean representations using t-SNE. Compared with deterministic embeddings, the proposed latent representation forms more compact intra-class clusters while maintaining clearer inter-class separation. Unknown identities are naturally distributed outside the enrolled identity clusters, indicating that the learned probabilistic embedding effectively preserves identity structure and enhances robustness for open-set biometric recognition.

4.3 Discussion

Performance Interpretation: The empirical results firmly validate the theoretical hypothesis: deterministic embeddings fail under severe unconstrained conditions because they are forced to encode observational noise into identity features. By representing palmprints as Gaussian distributions, our model successfully segregates the invariant identity traits (mean) from the structural noise (variance). Crucially, the integration of the U-Net Generative Decoder acts as a potent topological regularizer. It ensures that the probabilistic constraints do not aggressively destroy the intricate spatial patterns of palm creases, explaining why our full framework outperforms standard Probabilistic Face Embeddings (PFE) when adapted to palmprints.

Strengths and Robustness: A prominent strength of the proposed system is its exceptional capability in Open-Set Recognition (OSR). The system behaves as an intelligent attention and penalty mechanism. By thresholding the estimated uncertainty magnitude ($\boldsymbol{\sigma}$), the model proactively detects and rejects highly degraded inputs or out-of-distribution strangers. This dynamic rejection protocol significantly mitigates over-confident predictions, ensuring high system reliability in security-critical deployments. Additionally, discarding the generative decoder during inference guarantees that the computational overhead remains negligible, preserving real-time authentication speeds.

Limitations and Future Work: Despite the substantial improvements, certain limitations persist. Training the dual-pathway architecture (Encoder-Decoder) necessitates a meticulous progressive training strategy and careful hyper-parameter balancing (KL-annealing) to avoid initial instability. Furthermore, while the system flawlessly handles moderate blur and non-rigid deformations, extreme geometric distortions might still exceed the generative capacity of the regularizer. Future work will explore dynamic hierarchical uncertainty estimation and lightweight attention mechanisms to further stabilize the optimization of the generative pathway without requiring phased warm-ups.

5 Conclusion and Future Work

5.1 Conclusion

In this research, we introduced a novel uncertainty-aware framework tailored for unconstrained palmprint recognition. Recognizing the critical limitations of conventional deterministic embeddings when confronted with severe environmental noise and non-rigid geometric deformations, we transitioned to a probabilistic representation learning paradigm. By modeling each palmprint image as a multivariate Gaussian distribution, our system successfully disentangles the invariant identity characteristics (represented by the mean) from the observational data uncertainty (captured by the variance).

A core breakthrough of our methodology is the integration of a Generative Decoder, inspired by the Probabilistic U-Net, which acts exclusively as a topological regularizer during the training phase. This generative constraint effectively prevents the latent-space collapse typically induced by probabilistic regularization, ensuring the intricate geometric structures of palm creases are strictly preserved. During the inference phase, the decoder is entirely decoupled, enabling highly efficient, real-time biometric authentication.

Through comprehensive evaluations on unconstrained datasets, we demonstrated that the proposed probabilistic approach significantly mitigates the adverse effects of corrupted inputs, outperforming contemporary deterministic metric learning models. Furthermore, by tightly coupling the latent similarity matching with the estimated uncertainty bounds, our framework provides an exceptionally robust rejection mechanism. It proactively penalizes degraded samples and confidently rejects out-of-distribution strangers, thereby establishing a highly reliable and secure solution for Open-Set Recognition (OSR) deployments.

5.2 Future Work

While the proposed framework establishes a robust foundation for uncertainty-aware biometric identification, several promising avenues remain for future exploration:

- **Explicit Non-Rigid Deformation Modeling:** Although our generative regularizer preserves static spatial topologies, extreme geometric distortions caused by flexible hand articulations remain challenging. Future research could integrate explicit optical-flow-driven deformation modules (inspired by recent advancements in generative architectures like FlowPalm) directly into the regularization pathway. Simulating non-rigid structural transformations dynamically during training could further enhance the model's robustness to severe pose variations.
- **Holistic Uncertainty Estimation:** Currently, our framework primarily focuses on data uncertainty (aleatoric uncertainty) arising from input degradation. Transitioning toward a holistic uncertainty estimation paradigm which simultaneously quantifies data uncertainty and model uncertainty (epistemic uncertainty)could provide a more comprehensive probabilistic boundary. This holistic approach would further optimize the prediction rejection ratio when encountering highly anomalous adversarial attacks or entirely novel domains.
- **Hierarchical Multi-Layer Uncertainty:** The uncertainty of a palmprint image is influenced by both high-level semantic deformations and low-level textural noise. Future iterations of the architecture could explore hierarchical feature fusion mechanisms that extract and aggregate probabilistic distributions across multiple convolutional layers. This would allow the network to capture and penalize diverse sources of ambiguity more precisely.

- **Context-Aware Relational Matching:** Our current inference protocol evaluates similarity based on isolated pairwise probabilistic distances (such as FastMLS). To further suppress false acceptances in massive galleries, future work could incorporate context-aware relational reasoning. By evaluating how a pairwise similarity score compares within the broader context of the entire dataset distribution, the decision-making process would shift from absolute distance thresholding to dynamic, context-aware verification.

References

[1] Gao, C., Yang, Z., Jia, W., Leng, L., Zhang, B., Teoh, A.B.J.: Deep Learning in Palmprint Recognition-A Comprehensive Survey. arXiv (2025). https://doi.org/10.48550/arXiv.2501.01166

[2] Shi, Y., Jain, A.K.: Probabilistic Face Embeddings. arXiv (2019). https://doi.org/10.48550/arXiv.1904.09658

[3] Chang, J., Lan, Z., Cheng, C., Wei, Y.: Data Uncertainty Learning in Face Recognition. arXiv (2020). https://doi.org/10.48550/arXiv.2003.11339

[4] Erlygin, L., Zaytsev, A.: Holistic Uncertainty Estimation For Open-Set Recognition. arXiv (2024). https://doi.org/10.48550/ARXIV.2408.14229

[5] Hu, H., Wang, X., Zhang, Y., Chen, Q., Guan, Q.: A comprehensive survey on contrastive learning. Neurocomputing **610**, 128645 (2024) https://doi.org/10.1016/j.neucom.2024.128645

[6] Le-Khac, P.H., Healy, G., Smeaton, A.F.: Contrastive Representation Learning: A Framework and Review. IEEE Access **8**, 193907–193934 (2020) https://doi.org/10.1109/ACCESS.2020.3031549

[7] Kohl, S.A.A., Romera-Paredes, B., Meyer, C., Fauw, J.D., Ledsam, J.R., Maier-Hein, K.H., Eslami, S.M.A., Rezende, D.J., Ronneberger, O.: A Probabilistic U-Net for Segmentation of Ambiguous Images. arXiv (2019). https://doi.org/10.48550/arXiv.1806.05034

[8] Jin, J., Zhao, C., Zhang, R., Shang, S., Xu, J., Zhang, J., Wang, S., Zhao, Y., Ding, S., Jia, W., *et al.*: Diff-palm: Realistic palmprint generation with polynomial creases and intra-class variation controllable diffusion models. In: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 26367–26376 (2025)

[9] Lin, X., Duan, Y., Dong, Q., Lu, J., Zhou, J.: Deep Variational Metric Learning. In: Ferrari, V., Hebert, M., Sminchisescu, C., Weiss, Y. (eds.) Computer Vision – ECCV 2018 vol. 11219, pp. 714–729. Springer, Cham (2018). https://doi.org/10.1007/978-3-030-01267-0_42

[10] Turkoglu, M.O., Becker, A., Gündüz, H.A., Rezaei, M., Bischl, B., Daudt, R.C., D'Aronco, S., Wegner, J.D., Schindler, K.: FiLM-Ensemble: Probabilistic Deep Learning via Feature-wise Linear Modulation. arXiv (2022). https://doi.org/10.48550/arXiv.2206.00050

[11] Yang, Z., Huangfu, H., Leng, L., Zhang, B., Teoh, A.B.J., Zhang, Y.: Comprehensive Competition Mechanism in Palmprint Recognition. IEEE Transactions on Information Forensics and Security **18**, 5160–5170 (2023) https://doi.org/10.1109/TIFS.2023.3306104

[12] Genovese, A., Piuri, V., Plataniotis, K.N., Scotti, F.: PalmNet: Gabor-PCA Convolutional Networks for Touchless Palmprint Recognition. IEEE Transactions on Information Forensics and Security **14**(12), 3160–3174 (2019) https://doi.org/10.1109/TIFS.2019.2911165

[13] Shao, H., Zou, Y., Liu, C., Guo, Q., Zhong, D.: Learning to Generalize Unseen Dataset for Cross-Dataset Palmprint Recognition. IEEE Transactions on Information Forensics and Security **19**, 3788–3799 (2024) https://doi.org/10.1109/TIFS.2024.3371257

[14] Abid, A., Zou, J.: Contrastive Variational Autoencoder Enhances Salient Features. arXiv (2019). https://doi.org/10.48550/arXiv.1902.04601